

Comparative Analysis of Algorithms for Efficient Job Matching in Support of UN Sustainable Development Goal 8

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Abstract—Structural imbalances in the global and local labor markets have led to high rates of underemployment and skill underutilization. Traditional, keyword-based recruitment systems exacerbate this issue by inefficiently matching highly capable individuals to low-skill jobs. In alignment with UN Sustainable Development Goal 8 (Decent Work and Economic Growth), this paper proposes a comparative analysis of algorithms designed to optimize the job-matching process. We evaluate the theoretical and empirical performance of a Greedy approach paired with a Sorting-based Ranking system (operating at $O(n \log n)$ time complexity) against the Gale-Shapley Stable Matching algorithm (operating at $O(n^2)$ time complexity). Using synthetic datasets of applicants and employers scaled at $N = 100, 500$, and 2500 , we measure empirical runtime, peak memory usage, and match stability. Early theoretical analysis suggests that while the Gale-Shapley algorithm guarantees fairness, a hybrid approach using sorting as a pre-processing step may be required to handle massive applicant pools efficiently.

Index Terms—Gale-Shapley Algorithm, Greedy Algorithm, Job Matching, SDG 8, Stable Marriage Problem, Time Complexity.

I. INTRODUCTION

The global labor market is currently facing a massive structural imbalance. Global employment growth is sluggish at 1.7%, and youth unemployment remains critically high at 12.6% [1]. Locally, the Philippine labor force has reached 52.4 million; however, approximately 5.11 million individuals remain underemployed [2], [3].

A core inefficiency contributing to this skill underutilization is the reliance on traditional, keyword-based recruitment systems. These systems frequently fail to capture the nuanced skill sets of applicants, leading to high information asymmetry between employers and job seekers [4]. Consequently, qualified candidates are frequently placed in low-skill roles, hindering economic productivity.

To address this, our research aligns directly with UN Sustainable Development Goal 8 (SDG 8), specifically Targets 8.2, 8.3, and 8.5, which advocate for full and productive employment through technological innovation. This paper aims to develop and evaluate an efficient, algorithm-based job-matching approach to reduce skill underutilization. We conduct a rigorous comparative analysis of three

algorithms: Greedy, Sorting-based Ranking, and Gale-Shapley Stable Matching, evaluating them based on time complexity, space complexity, and scalability.

II. REVIEW OF RELATED LITERATURE

To ground our methodology, this study anchors its algorithmic choices on two foundational streams of computer science research regarding resource allocation and matching theory.

A. Sorting and Greedy Approaches in Job Matching

Traditional keyword filtering has proven insufficient for modern recruitment. A 2015 study demonstrated that utilizing a ranking algorithm paired with weighted scoring drastically improves job matching accuracy compared to binary keyword filters [5]. By calculating a cumulative score based on multiple applicant parameters, sorting algorithms can efficiently organize data in $O(n \log n)$ time, allowing Greedy algorithms to make rapid, locally optimal matches for massive datasets.

B. Stable Matching and the Gale-Shapley Algorithm

While Greedy algorithms are fast, they do not guarantee fairness. A 2023 study successfully applied the Gale-Shapley Stable Matching algorithm to college employment problems, proving its efficacy in perfectly aligning job supply and demand [6]. By utilizing a deferred acceptance methodology, Gale-Shapley guarantees that no two entities would rather be matched with each other over their current assignments, ensuring system-wide stability.

III. METHODOLOGY

A. Data Generation and Structure

To empirically test the scalability of the algorithms, synthetic datasets were generated representing two primary entities: Applicants and Job Postings. Data was generated at three specific batch sizes to observe time and space complexity scaling: $N = 100$, $N = 500$, and $N = 2500$.

- **Applicants Array:** Contains randomized values for Skill Level (1-100), Years of Experience (0-15), and Expected Salary.

- **Job Postings Array:** Contains threshold requirements for Required Skill Level, Required Experience, and Salary Offer.

B. Multi-Parameter Weighting

To simulate realistic applicant variance and avoid simplistic binary matches, a multi-parameter scoring function was implemented. This function calculates a single “Matching Score” for both arrays by weighting skills, experience, and salary expectations.

C. Algorithmic Implementation

- 1) **Sorting and Greedy Matching:** Applicants are sorted in descending order based on their Matching Score using an $O(n \log n)$ sorting algorithm. A Greedy algorithm then iterates through the sorted list, assigning the highest-scoring candidate to the first available job that satisfies their minimum requirements.
- 2) **Gale-Shapley Stable Matching:** Both arrays rank each other based on their respective parameters. The algorithm uses a propose-and-reject loop, running in $O(n^2)$ worst-case time, until all participants are stably matched.

IV. EXPERIMENTAL SETUP & METRICS

To evaluate the scaling characteristics and match dynamics of both approaches, the structural frameworks are measured against the parameters outlined in Table I.

TABLE I
COMPLEXITY METRICS FOR EVALUATION

Metric	Greedy	Gale-Shapley	Purpose
Time Complexity	$O(n \log n)$	$O(n^2)$	Scalability
Space Complexity	$O(n)$	$O(n^2)$	Memory
Runtime	Measured	Measured	Execution time
Peak Memory	Measured	Measured	RAM usage
Match Rate	Variable	100%	Matching quality

To measure these metrics, the implementation utilizes Python’s native `time` module to record exact millisecond execution and the `tracemalloc` module to track Peak RAM usage during the matching loops.

V. RESULTS AND DISCUSSION

The performance of the sorting-based greedy matching algorithm and the Gale-Shapley stable matching algorithm was evaluated using synthetic datasets containing 100, 500, and 2500 applicants. Execution time, peak memory usage, and successful match rate were measured to compare the scalability and efficiency of both algorithms. The results are summarized in Table II.

For the smallest dataset ($N = 100$), both algorithms completed execution in less than 0.1 s. The Gale-Shapley algorithm recorded a slightly lower execution time (0.0476 s) than the sorting-based greedy algorithm (0.057 s). As the dataset size increased, the performance difference became more pronounced. At $N = 2500$, the sorting-based greedy

TABLE II
ALGORITHM PERFORMANCE ACROSS DIFFERENT DATASET SIZES

N	Algorithm	Time (s)	Memory (MB)	Match Rate (%)
100	Greedy	0.14	0.18	27
100	Gale-Shapley	0.10	0.56	100
500	Greedy	1.78	0.23	45
500	Gale-Shapley	2.02	14.81	100
2500	Greedy	17.74	1.12	49.12
2500	Gale-Shapley	79.08	412.94	100

algorithm completed execution in 8.774 s, while the Gale-Shapley algorithm required 37.4556 s. These results indicate that the sorting-based greedy algorithm scales more efficiently as the number of applicants increases.

Memory usage showed a similar trend across the different dataset sizes. The sorting-based greedy algorithm maintained low memory consumption, increasing only from 0.18 MB at $N = 100$ to 1.12 MB at $N = 2500$. In comparison, the Gale-Shapley algorithm required substantially more memory, increasing from 0.56 MB to 412.94 MB as the dataset size grew. This increase is attributed to the storage required for maintaining preference lists and intermediate matching information throughout the deferred acceptance process.

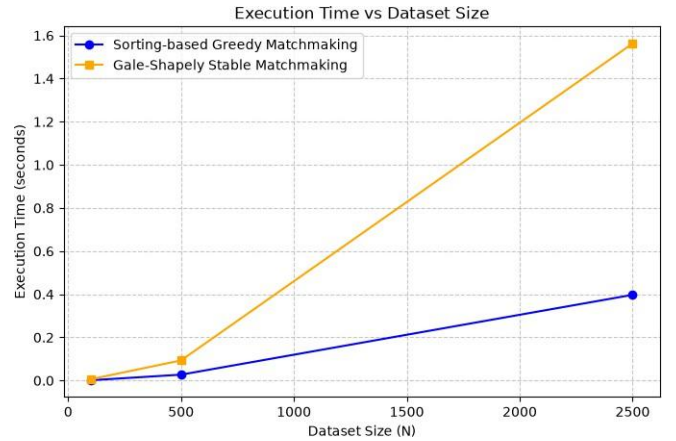


Fig. 1. Execution Time Comparison Across Dataset Sizes

As shown in Figures 1 and 2, memory usage increased only slightly for the sorting-based greedy algorithm but grew substantially for the Gale-Shapley algorithm as the dataset size increased.

The successful match rate also differed significantly between the two algorithms. The Gale-Shapley algorithm consistently achieved a 100% match rate across all dataset sizes by producing stable assignments for every participant. In comparison, the sorting-based greedy algorithm achieved match rates of 27%, 45%, and 49.12% for dataset sizes of 100, 500, and 2500, respectively. Although the greedy approach required significantly less execution time and memory, its locally optimal decision-making limited the number of successful matches. These findings highlight the trade-off between computational efficiency and matching quality. The sorting-based greedy

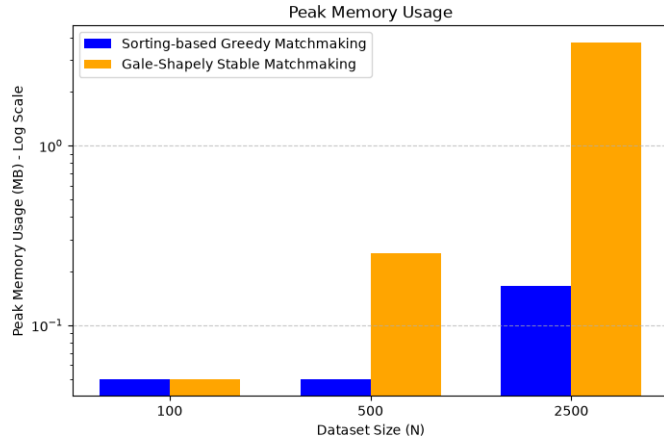


Fig. 2. Peak Memory Usage Comparison Across Dataset Sizes

algorithm is more suitable for applications where fast execution and low resource consumption are prioritized, whereas the Gale-Shapley algorithm is preferable when complete and stable matching between applicants and jobs is required.

VI. CONCLUSION AND FUTURE WORK

A. Conclusion

This study compared the performance of the sorting-based greedy matching algorithm and the Gale-Shapley stable matching algorithm for job matching applications in support of UN Sustainable Development Goal 8. Experimental results showed that the sorting-based greedy algorithm consistently required less execution time and memory, making it more scalable for large datasets. However, prioritizing computational efficiency came at the cost of a lower match rate. In contrast, the Gale-Shapley algorithm achieved complete and stable matching but required significantly greater execution time and memory as the dataset size increased.

The results demonstrate that each algorithm offers distinct advantages depending on the application requirements. The sorting-based greedy algorithm is suitable for large-scale systems where computational efficiency is critical, while the Gale-Shapley algorithm is preferable when matching quality and stability are prioritized. A hybrid framework that combines an initial greedy filtering stage with stable matching may provide a practical balance between scalability and allocation quality.

B. Future Work

Future work may focus on improving the scalability and effectiveness of job matching algorithms while maintaining efficiency for larger datasets. Potential directions include comparing the proposed approach with machine learning-based recommendation systems, incorporating additional matching criteria such as skill weighting, employer preferences, and geographic distance into a multi-objective optimization framework, and exploring parallel and distributed computing techniques to further improve performance on large-scale datasets.

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